Análisis Histórico de la Copa del Mundo FIFA (1930–2018)

Proyecto Individual — COM4602

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Contents

1 Descripción del proyecto

El proyecto analiza datos históricos de la Copa del Mundo FIFA (1930–2018) obtenidos desde TidyTuesday. Se aplican técnicas de análisis exploratorio y modelado predictivo multinomial/logístico para predecir el resultado de un partido (`outcome`: Home/Away/Draw).

2 Carga y limpieza de datos

```
source("R/load.R")
```

```
## Rows: 21
## Columns: 10
## $ year         <int> 1930, 1934, 1938, 1950, 1954, 1958, 1962, 1966, 1970, 197~
## $ host         <chr> "Uruguay", "Italy", "France", "Brazil", "Switzerland", "S~
## $ winner       <chr> "Uruguay", "Italy", "Italy", "Uruguay", "West Germany", "~
## $ second       <chr> "Argentina", "Czechoslovakia", "Hungary", "Brazil", "Hung~
## $ third        <chr> "USA", "Germany", "Brazil", "Sweden", "Austria", "France"~
## $ fourth       <chr> "Yugoslavia", "Austria", "Sweden", "Spain", "Uruguay", "W~
## $ goals_scored <int> 70, 70, 84, 88, 140, 126, 89, 89, 95, 97, 102, 146, 132, ~
## $ teams        <int> 13, 16, 15, 13, 16, 16, 16, 16, 16, 16, 16, 24, 24, 24, 2~
## $ games        <int> 18, 17, 18, 22, 26, 35, 32, 32, 32, 38, 38, 52, 52, 52, 5~
## $ attendance   <int> 434000, 395000, 483000, 1337000, 943000, 868000, 776000, ~
## Rows: 900
## Columns: 15
## $ year           <int> 1930, 1930, 1930, 1930, 1930, 1930, 1930, 1930, 1930, 1~
## $ country        <chr> "Uruguay", "Uruguay", "Uruguay", "Uruguay", "Uruguay", ~
## $ city           <chr> "Montevideo", "Montevideo", "Montevideo", "Montevideo",~
## $ stage          <chr> "Group 1", "Group 4", "Group 2", "Group 3", "Group 1", ~
## $ home_team      <chr> "France", "Belgium", "Brazil", "Peru", "Argentina", "Ch~
## $ away_team      <chr> "Mexico", "United States", "Yugoslavia", "Romania", "Fr~
## $ home_score     <int> 4, 0, 1, 1, 1, 3, 0, 0, 1, 6, 1, 0, 0, 4, 3, 6, 6, 4, 2~
## $ away_score     <int> 1, 3, 2, 3, 0, 0, 4, 3, 0, 3, 0, 1, 4, 0, 1, 1, 1, 2, 3~
## $ outcome        <chr> "H", "A", "A", "A", "H", "H", "A", "A", "H", "H", "H", ~
## $ win_conditions <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "",~
## $ winning_team   <chr> "France", "United States", "Yugoslavia", "Romania", "Ar~
## $ losing_team    <chr> "Mexico", "Belgium", "Brazil", "Peru", "France", "Mexic~
## $ date           <chr> "1930-07-13", "1930-07-13", "1930-07-14", "1930-07-14",~
## $ month          <chr> "Jul", "Jul", "Jul", "Jul", "Jul", "Jul", "Jul", "Jul",~
## $ dayofweek      <chr> "Sunday", "Sunday", "Monday", "Monday", "Tuesday", "Wed~
## worldcups: 21 filas; 10 columnas.
## wcmatches: 900 filas; 15 columnas.
```

```r
source("R/clean.R")
```

2.1 Datos cargados

```r
glimpse(worldcups)
```

```
## Rows: 21
## Columns: 12
## $ year                <int> 1930, 1934, 1938, 1950, 1954, 1958, 1962, 1966, 19~
## $ host                <chr> "Uruguay", "Italy", "France", "Brazil", "Switzerla~
## $ winner              <chr> "Uruguay", "Italy", "Italy", "Uruguay", "West Germ~
## $ second              <chr> "Argentina", "Czechoslovakia", "Hungary", "Brazil"~
## $ third               <chr> "USA", "Germany", "Brazil", "Sweden", "Austria", "~
## $ fourth              <chr> "Yugoslavia", "Austria", "Sweden", "Spain", "Urugu~
## $ goals_scored        <int> 70, 70, 84, 88, 140, 126, 89, 89, 95, 97, 102, 146~
## $ teams               <int> 13, 16, 15, 13, 16, 16, 16, 16, 16, 16, 16, 24, 24~
## $ games               <int> 18, 17, 18, 22, 26, 35, 32, 32, 32, 38, 38, 52, 52~
## $ attendance          <int> 434000, 395000, 483000, 1337000, 943000, 868000, 7~
## $ goals_per_game      <dbl> 3.888889, 4.117647, 4.666667, 4.000000, 5.384615, ~
## $ attendance_per_game <dbl> 24111.11, 23235.29, 26833.33, 60772.73, 36269.23, ~
```

```r
glimpse(wcmatches)
```

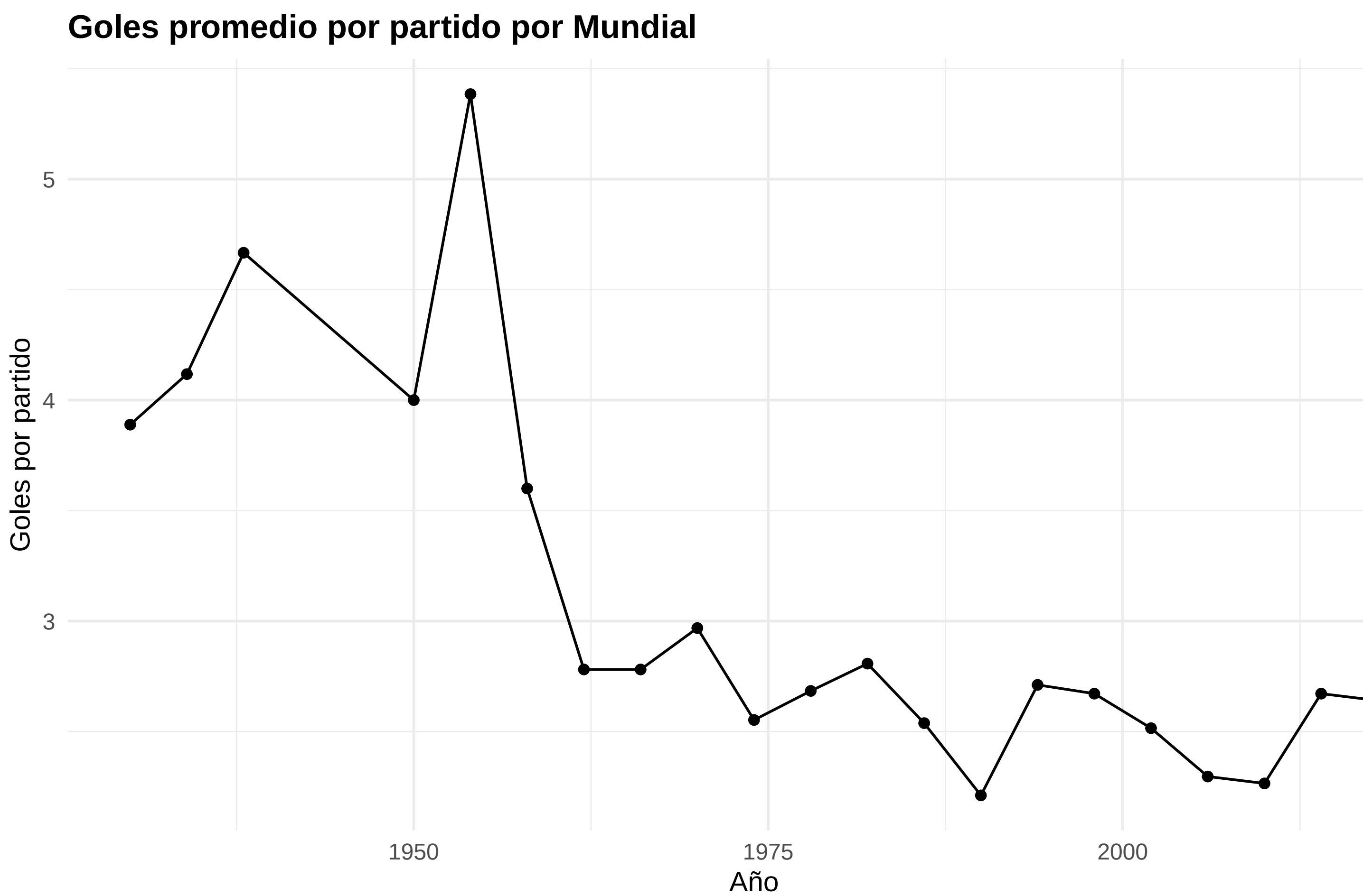
```
## Rows: 900
## Columns: 18
## $ year            <int> 1930, 1930, 1930, 1930, 1930, 1930, 1930, 1930, 1930, ~
## $ country         <chr> "Uruguay", "Uruguay", "Uruguay", "Uruguay", "Uruguay",~
## $ city            <chr> "Montevideo", "Montevideo", "Montevideo", "Montevideo"~
## $ stage           <chr> "Group 1", "Group 4", "Group 2", "Group 3", "Group 1",~
## $ home_team       <chr> "France", "Belgium", "Brazil", "Peru", "Argentina", "C~
## $ away_team       <chr> "Mexico", "United States", "Yugoslavia", "Romania", "F~
## $ home_score      <int> 4, 0, 1, 1, 1, 3, 0, 0, 1, 6, 1, 0, 0, 4, 3, 6, 6, 4, ~
## $ away_score      <int> 1, 3, 2, 3, 0, 0, 4, 3, 0, 3, 0, 1, 4, 0, 1, 1, 1, 2, ~
## $ outcome         <chr> "H", "A", "A", "A", "H", "H", "A", "A", "H", "H", "H",~
## $ win_conditions  <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", ""~
## $ winning_team    <chr> "France", "United States", "Yugoslavia", "Romania", "A~
## $ losing_team     <chr> "Mexico", "Belgium", "Brazil", "Peru", "France", "Mexi~
## $ date            <chr> "1930-07-13", "1930-07-13", "1930-07-14", "1930-07-14"~
## $ month           <chr> "Jul", "Jul", "Jul", "Jul", "Jul", "Jul", "Jul", "Jul"~
## $ dayofweek       <chr> "Sunday", "Sunday", "Monday", "Monday", "Tuesday", "We~
## $ goals_per_match <int> 5, 3, 3, 4, 1, 3, 4, 3, 1, 9, 1, 1, 4, 4, 4, 7, 7, 6, ~
## $ empate          <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
## $ stage_type      <chr> "Group Stage", "Group Stage", "Group Stage", "Group St~
```

3 Análisis exploratorio

```r
source("R/eda.R")
```

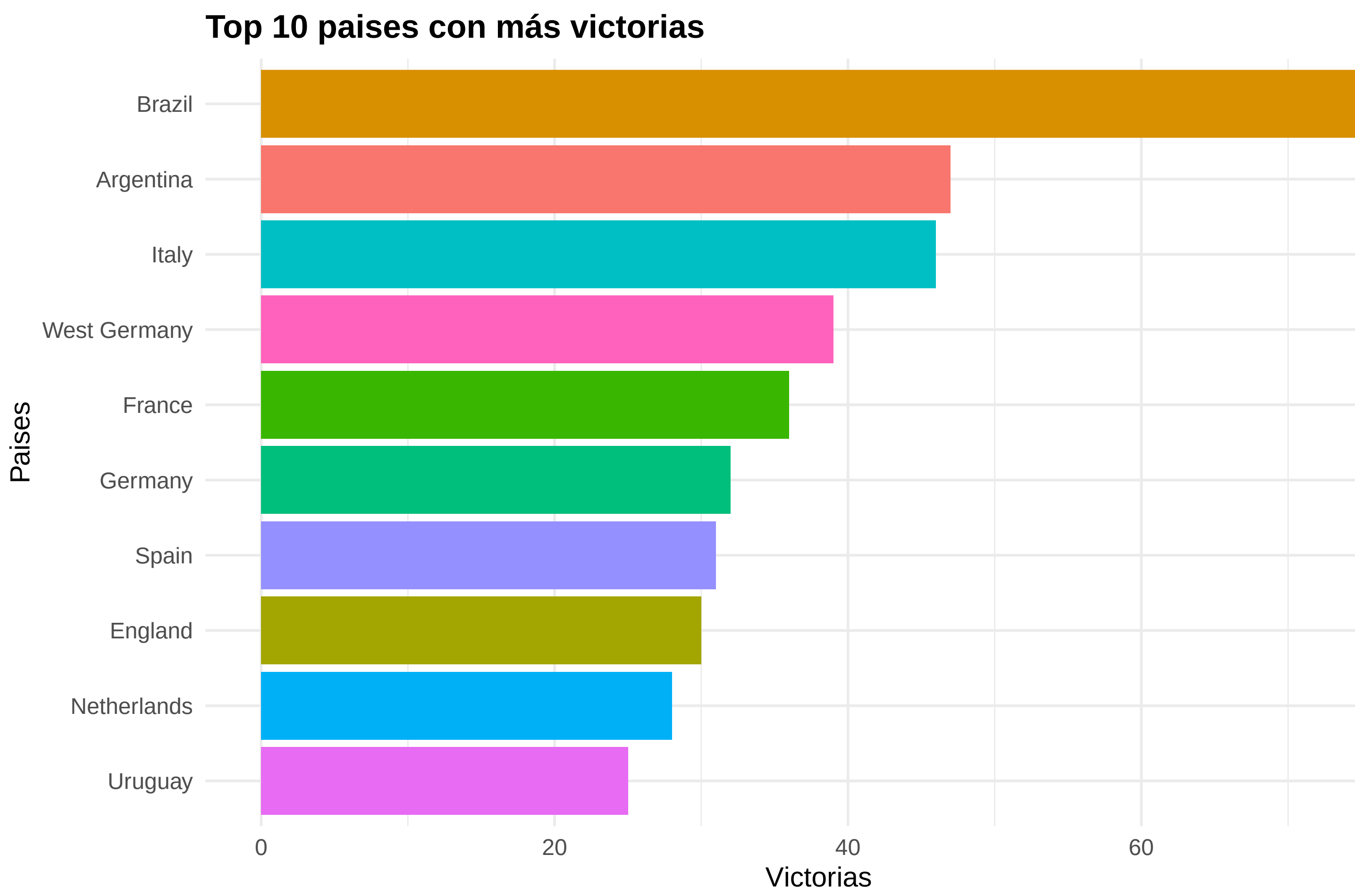
3.1 Evolución de goles por torneo

```r
p1
```



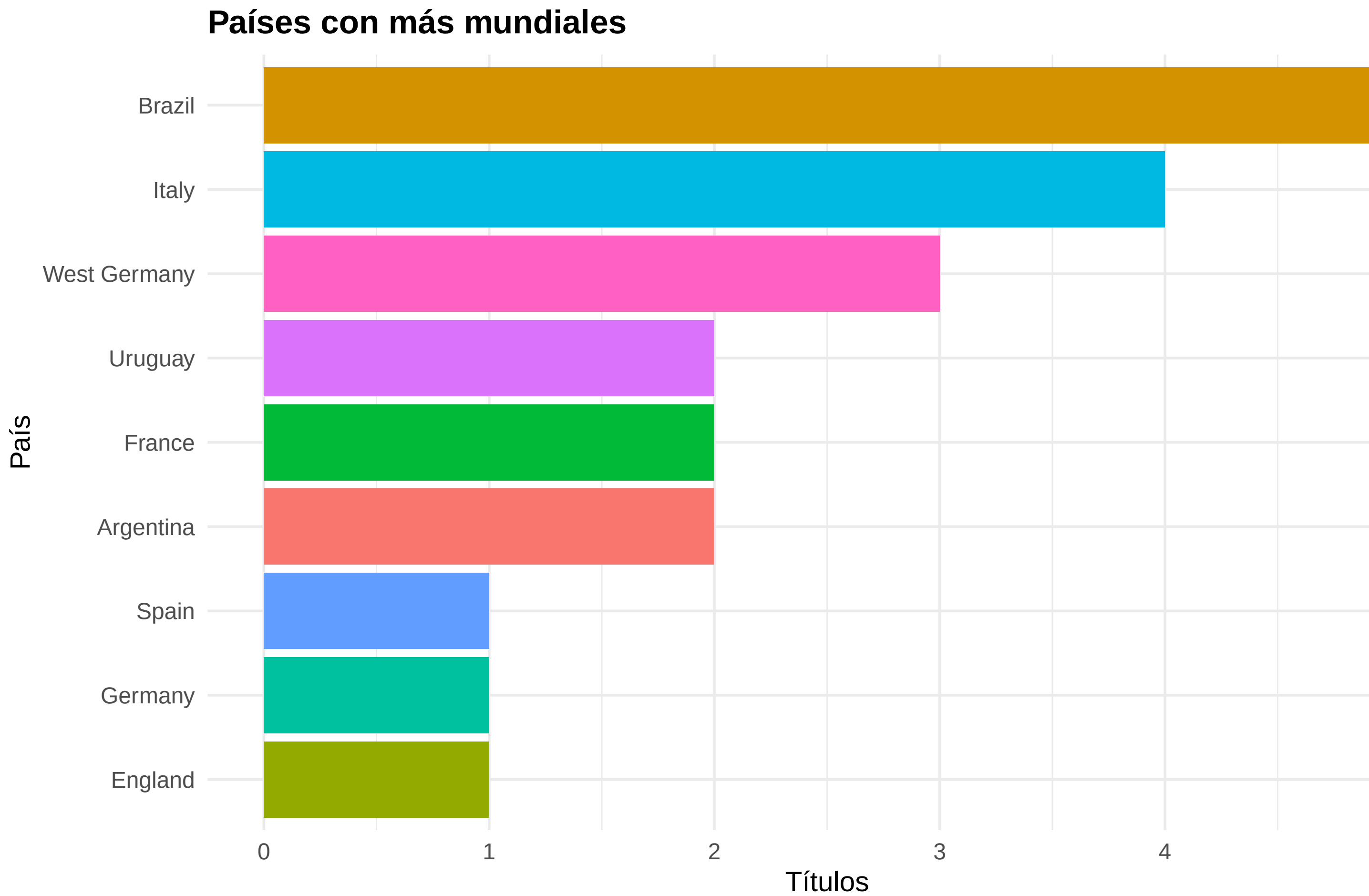
3.2 Top 10 selecciones con más victorias

```
p2
```



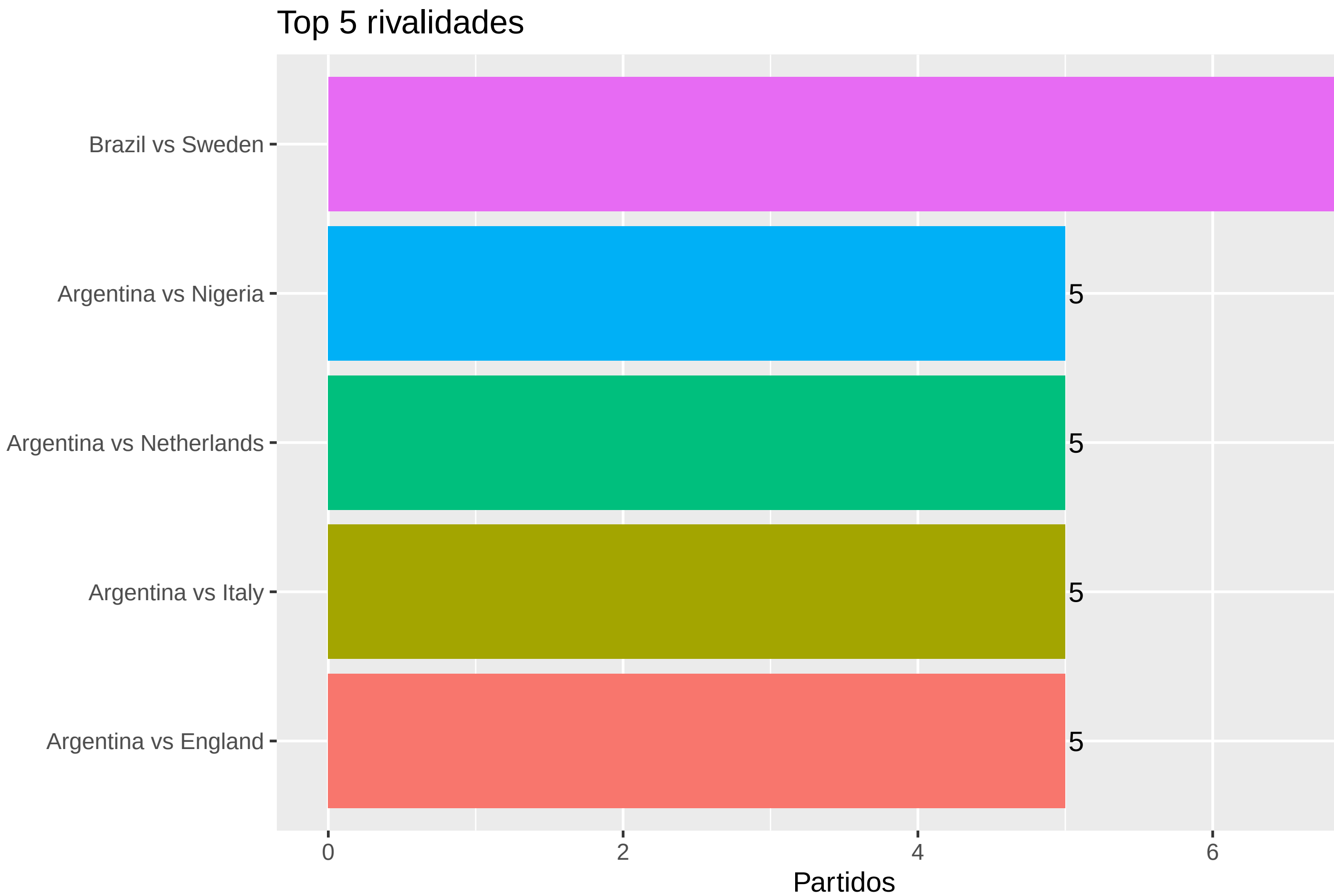
3.3 Países con más títulos

```
p3
```



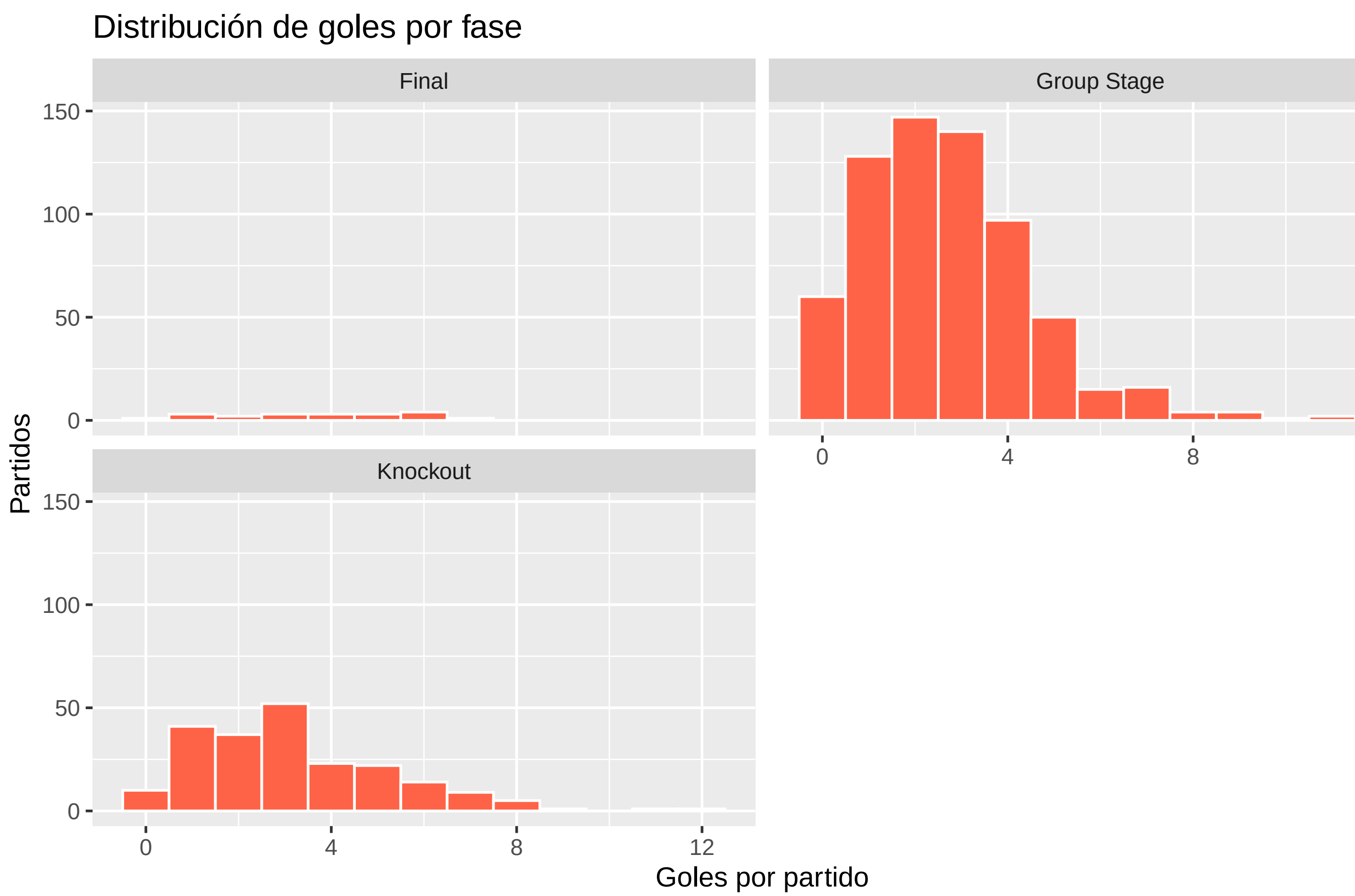
3.4 Top 5 rivalidades

```
p4
```



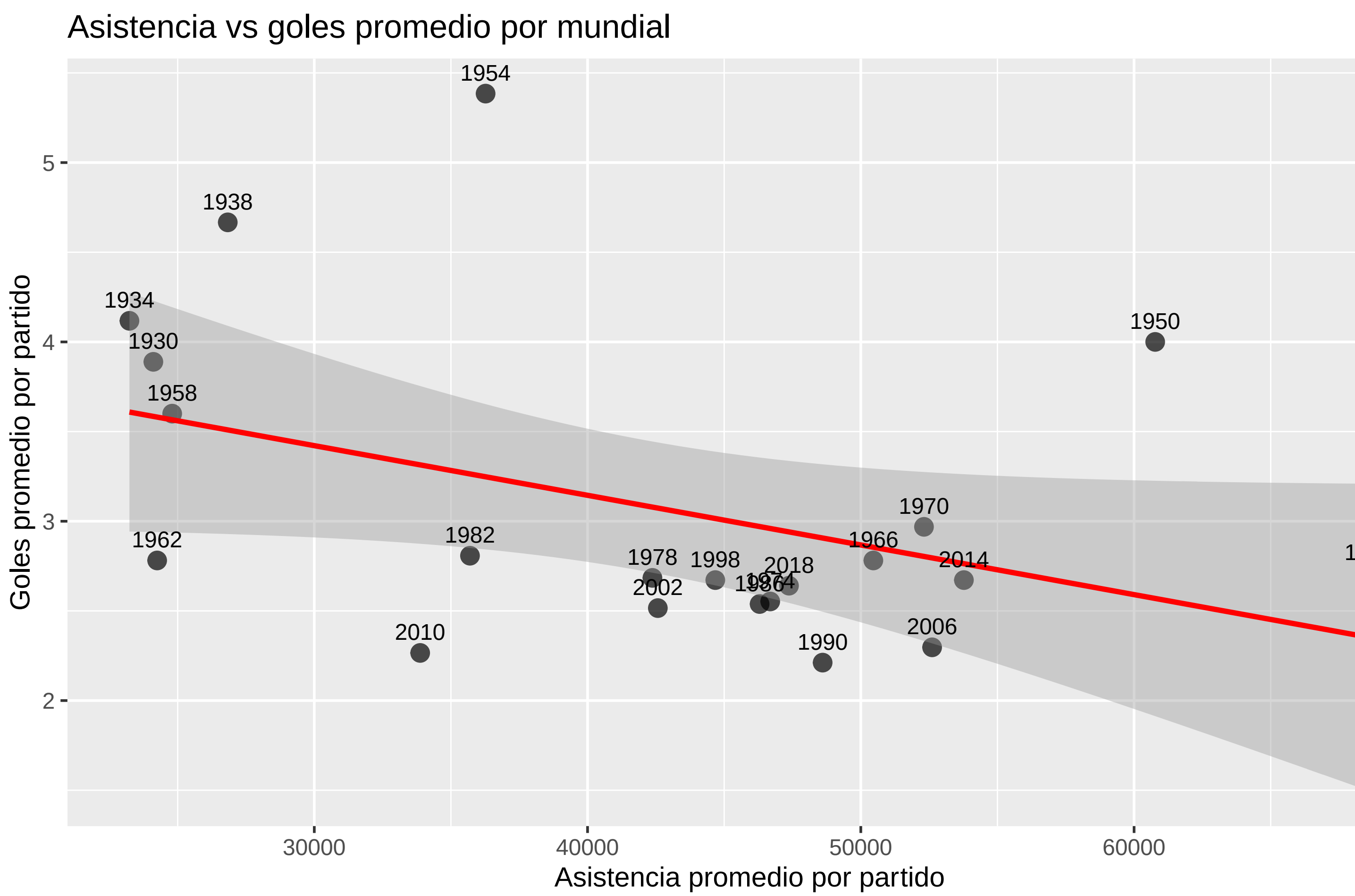
3.5 Distribución de goles por fase

```
p5
```



3.6 Asistencia vs goles promedio por mundial

```
p6
```



4 Modelado predictivo

Se predice `outcome` (H/A/D) usando regresión logística multinomial. Se comparan tres enfoques: StepAIC, Ridge y Lasso.

```
source("R/modeling.R")
```

4.1 Partición de datos

- **Train:** 70% (630 observaciones)
- **Test:** 30% (270 observaciones)

```
table(train$outcome)
```

```
##
##   A   D   H
## 206 122 302
```

```r
table(test$outcome)
```

```
##
##   A   D   H
##  96  47 127
```

```r
prop.table(table(train$outcome)) %>% round(3)
```

```
##
##     A     D     H
## 0.327 0.194 0.479
```

```r
prop.table(table(test$outcome)) %>% round(3)
```

```
##
##     A     D     H
## 0.356 0.174 0.470
```

La clase dominante es `H` (victoria local), reflejando ventaja de local histórica en mundiales. Las clases están distribuidas de forma similar entre train y test, lo que valida la partición.

4.2 Resultados

```r
acc
```

```
##    Modelo Accuracy
## 1 StepAIC   97.78%
## 2   Ridge    93.7%
## 3   Lasso   97.41%
```

El modelo con mayor accuracy es `StepAIC`. Ridge y Lasso aplican regularización sobre los coeficientes del multinomial, penalizando variables poco informativas. StepAIC selecciona variables por criterio AIC desde el modelo nulo. La variable `home_score` y `away_score` son los predictores más influyentes dado que determinan directamente el outcome.